Lab 7 - Fuzz Testing Deliverables

Scholarship Awards | Hypothesis input-validation tests

Deliverables included

1. This PDF: original output, failing fuzz tests, fixed output, and passing tests.
2. scholarship_awards.py: hardened application; malformed input is handled without crashing.
3. test_scholarship_awards.py: Hypothesis fuzz, valid-input, and invalid-input regression tests.

Verification: fixed suite passes all 5 tests; sample results remain Merit/82.0, 82, and 5400.0.

The pre-fix fuzz run reproduces failures in all three functions.

Evidence panels render captured output from the listed commands.

STEP 2 - Original application output

PowerShell - Lab 7 console evidence

```
PS> python scholarship_awards.py

==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

Exit code: 0

STEP 4 - Hypothesis failures before fix

PowerShell - Lab 7 console evidence

```
PS> python -m pytest -q --tb=no test_scholarship_awards.py

FFF
                                                                        [100%]
=============================================================================== short test summary info
    ===========================================================================
FAILED test_scholarship_awards.py::test_parse_applicant_no_crash - ExceptionGroup: Hypothesis found 3 distinct
    failures. (3 sub-exceptions)
FAILED test_scholarship_awards.py::test_marks_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct
    failures. (2 sub-exceptions)
FAILED test_scholarship_awards.py::test_payable_award_no_crash - ExceptionGroup: Hypothesis found 2 distinct
    failures. (2 sub-exceptions)
3 failed in 4.85s
```

Exit code: 1 (expected failures)

STEP 5 - Fixed application output

PowerShell - Lab 7 console evidence

```
PS> python scholarship_awards.py

==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

Exit code: 0

STEP 6 - Fuzz tests after fix

